Jack Ngo

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Education

University of North Texas Denton, TX
Bachelor of Science in Computer Science; GPA: 3.9/4.0 *Aug. 2024 – May 2028*
- Relevant coursework: Data Structures & Algorithms, System Programming, Internet Programming, Database Systems, Software Engineering, and Linear Algebra

Experience

Software Engineering Intern Remote
Eudaimonic Inc *Feb. 2026 – May 2026*
- Shipped **3+ full-stack React applications** with live multi-user synchronization and presence, **60 FPS game loops**, and drift-free sequencer timing on a **25 ms scheduler tick** using React, TypeScript, Web Audio API, Express, and MongoDB
- Cut storage writes by an estimated **70%** by debouncing autosave, keeping transport state client-side, and computing progression state on read with React, TypeScript, and a SQLite-backed record store
- Shipped an **AI slide-generating agent** that creates and edits live React slides by building a two-stage planner-builder pipeline with the Vercel AI SDK, Zod, and Anthropic/OpenAI models
- Achieved **100% compile success across a 5,000-slide benchmark** by building a browser-side TSX-to-React compiler using React, TypeScript, and Sucrase
- Prevented **uncapped AI spending** while retaining no-signup trial access by designing a server-side, token-scoped billing architecture with Cloudflare Workers, Hono, and JWT authentication

Undergraduate Research Assistant and Developer University of North Texas
High Performance Cloud Computing Lab and AI Summer Research *May 2025 – Dec. 2025*
- Developed a **multi-threaded Piper TTS engine** with hierarchical message queues for real-time audio on SmartSight glasses
- Validated real-time audio systems end-to-end through **automated test suites** and keyboard-triggered threads
- Developed an end-to-end **Python UAV imagery pipeline** with geography researchers to automate *Ligustrum sinense* detection, replacing manual field surveys and driving **$1M+ in cost avoidance**
- Trained YOLO and Faster R-CNN with a ResNet-101 backbone to reach an **F1 score of up to 91%** for *Ligustrum sinense* classification
- Authored unit tests for a YOLOv8 annotation tool and **auto-labeled 600+ images** to reduce manual labeling effort

Projects

Document Extraction Platform | *React, Express, SQLite, LLMs, sqlite-vec, ChromaDB* Jul. 2026 – Present
- Built a **multi-tenant document-extraction platform** using Ollama Qwen to extract structured data from files, reducing manual data-entry time by **60–70%**
- Designed a **three-layer hallucination-detection pipeline** with server-side verbatim quote matching, an isolated grounding check, and confidence scoring, catching **100% of fabricated quotes** before human review
- Implemented a **RAG pipeline** with provenance-preserving parsing, vector embeddings, and reciprocal rank fusion, enabling natural-language queries with **server-verified citations** traceable to exact source locations

Agentic RAG Support Bot | *Python, LangGraph, FastAPI, ChromaDB, BM25* Jul. 2026
- Built a bilingual Vietnamese/English **RAG support agent** over **250 help-center pages** for an education-management SaaS using Gemini, hybrid dense + BM25 retrieval, and cross-encoder reranking to reach **0.97 Recall@10**
- Designed a **self-correcting LangGraph agent** that retries retrieval up to three times and routes low-confidence answers to human handoff instead of guessing, reaching **1.00 faithfulness with zero hallucinations**
- Built a **39-question evaluation harness** for retrieval recall and answer faithfulness, validating **87% tier-1 ticket deflection** at **under 2 seconds response time**

Technical Skills

Languages: Python, TypeScript, JavaScript, C/C++ (OOP), SQL
Frameworks & Libraries: React, Node.js, Express, FastAPI, LangGraph, YOLO
Databases & Storage: SQLite, MongoDB, ChromaDB
Tools & Technologies: Git, Linux, WebSocket, Tesseract OCR